**DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS**

# Mathematics II Advanced Calculus (2AC)

**Tutorial Sheet 13:** **18th and 19th May 2023**

1. We first summarize in this section, some of the main identities established or used in the lectures. Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a continuous function and $\vec{F} : \mathbb{R}^3 \to \mathbb{R}^3$ be a (continuous) vector field. Let $\mathcal{C}$ be a regular curve described by a vecor function $\vec{r} : [a, b] \to \mathbb{R}^3$. Then we recall that

$$\left\{\begin{array}{l}
\underbrace{\int_{\mathcal{C}} f(x,y,z)ds = \int_a^b f(\vec{r}(t))\ \|\vec{r}\,'(t)\|\ dt\,,}_{\text{Line integral of scalar-valued function}} \qquad \underbrace{\int_{\mathcal{C}} \vec{F}\cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t))\cdot\vec{r}\,'(t)\,dt}_{\text{Line integrals of a vector field}} \\[2ex]
\underbrace{\int_{\mathcal{C}} \nabla f\cdot d\vec{r} = \int_a^b \nabla f(\vec{r}(t))\cdot\vec{r}\,'(t)\,dt = \int_a^b \frac{d(f\circ\vec{r})}{dt}\,dt = f(\vec{r}(b)) - f(\vec{r}(a))\,,}_{(f \text{ of class } C^1!\ \text{Fundamental Theorem of Calculus for line integrals})} \\[2ex]
\underbrace{\int_{\mathcal{C}} = \sum_{i=1}^{n}\int_{\mathcal{C}_i}}_{c \text{ piecewise regular}} \quad \text{when } \mathcal{C} = \bigcup_{i=1}^{n}\mathcal{C}_i \text{ with } \mathcal{C}_i\cap\mathcal{C}_j = \{\text{single point at most}\} \text{ for } i\neq j \\[2ex]
(*)\ \int_{-\mathcal{C}} f ds = \int_{\mathcal{C}} f ds \quad \text{while} \quad (**)\ \int_{-\mathcal{C}} \vec{F}\cdot d\vec{r} = -\int_{\mathcal{C}} \vec{F}\cdot d\vec{r} \\
-\mathcal{C} \text{ denotes the curve having the same points as } \mathcal{C} \text{ with the opposite orientation}
\end{array}\right.$$

We want to explain why $(*)$ and $(**)$ hold for a regular curve $\mathcal{C}$ in $\mathbb{R}^2$ and $f : \mathbb{R}^2 \to \mathbb{R}$).

(1) Let $\vec{r} : [a, b] \to \mathbb{R}^2$ be a vector function describing $\mathcal{C}$ and deduce a vector function $\vec{u} : [a, b] \to \mathbb{R}^2$ describing the curve $-\mathcal{C}$.

(2) Use the vector functions $\vec{r}$ and $\vec{u}$ to evaluate $\displaystyle\int_{\mathcal{C}} f(x,y)ds$ and $\displaystyle\int_{-\mathcal{C}} f(x,y)ds$ respectively and check that $(*)$ holds.

(3) Use the vector functions $\vec{r}$ and $\vec{u}$ to evaluate $\displaystyle\int_{\mathcal{C}} \vec{F}\cdot d\vec{r}$ and $\displaystyle\int_{-\mathcal{C}} \vec{F}\cdot d\vec{r}$ respectively and check that $(**)$ holds.

2. Let $\vec{F} : \mathbb{R}^2 \to \mathbb{R}^2$ be the vector field defined by $\vec{F}(x, y) = (2x^2y^3, 3x^2y^2)$ for every $(x, y) \in \mathbb{R}^2$.

   (a) Use the Green theorem to evaluate the line integral $\oint_C \vec{F} \cdot d\vec{r}$ where $C$ is the piecewise regular curve in the first quadrant with an anticlockwise orientation, formed by the $x$-axis, the line $x = 1$ and and the curve $y = x^2$.

   (b) Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ directly. (Hint: Use the formula for the line integral on a piecewise regular curve. So, write $C = C_1 \cup C_2 \cup C_3$, and find a suitable vector function for each arc of curve $C_1, C_2, C_3$ taking into account the fact that $C$ has a counterclockwise orientation.)

3. If $\vec{F} = (P, Q) : \mathbb{R}^2 \to \mathbb{R}^2$ is a $\mathcal{C}^1$ vector field such that $P_y(x, y) = Q_x(x, y)$ in a subset $U$ of $\mathbb{R}^2$, then the line integral
$$\oint_{\mathcal{C}} P(x, y)dx + Q(x, y)dy = 0$$
for any piecewise regular simple closed curve $\mathcal{C}$ contained in $U$.
Hint: Consider both cases: (i) $\mathcal{C}$ with a counterclockwise orientation and (ii) $\mathcal{C}$ with a clockwise orientation.

4. We first summarize in this section, some of the main identities established or used in the lectures. Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a continuous function and $\vec{F} : \mathbb{R}^3 \to \mathbb{R}^3$ be a (continuous) vector field. Let $\mathcal{S}$ be a regular surface described by the vector function $\vec{r} : D \subset \mathbb{R}^2 \to \mathbb{R}^3$. Then,

$$\underbrace{A(\mathcal{S}) = \iint_D \|\vec{r}_s(s,t) \times \vec{r}_t(s,t)\| \, dsdt}_{\text{The area of the surface } \mathcal{S}}$$

$$\underbrace{\iint_{\mathcal{S}} f(x,y,z)dS = \iint_D f(\vec{r}(s,t)) \, \|\vec{r}_s(s,t) \times \vec{r}_t(s,t)\| \, dsdt}_{\text{The surface integral of the function } f \text{ over the surface } \mathcal{S}}$$

$$\underbrace{\iint_{\mathcal{S}} \vec{F}(x,y,z) \cdot \vec{n}dS = \iint_D \vec{F}(\vec{r}(s,t)) \cdot [\vec{r}_s(s,t) \times \vec{r}_t(s,t)] \, dsdt}_{\text{The flux integral of the vector field } \vec{F} \text{ across the surface } \mathcal{S}}$$

$$\underbrace{\iint_{\mathcal{S}} f(x,y,z)dS = \sum_{i=1}^{n} \iint_{\mathcal{S}_i} f(x,y,z)dS \quad \text{and} \quad \iint_{\mathcal{S}} \vec{F} \cdot \vec{n}dS = \sum_{i=1}^{n} \iint_{\mathcal{S}_i} \vec{F} \cdot \vec{n}dS}_{\text{In case } \mathcal{S} \text{ is piecewise regular: } \mathcal{S} = \bigcup_{i=1}^n \mathcal{S}_i \text{ with each } \mathcal{S}_i \text{ regular and } \mathcal{S}_\rangle \cap \mathcal{S}_j \subset \{\text{a curve}\} \text{ for } i \neq j}$$

(a) Show that $\iint_S xy\,dS = 4\int_0^1\int_0^{\frac{\pi}{2}} r^3\cos\theta\sin\theta\sqrt{r^2(4+5\cos^2\theta)+1}\,d\theta dr$ where the surface $S$ described by the vector function $\vec{r}(s,t)=(s,2t,s^2+3t^2)$ with $s^2+t^2\leq 1$, $s\geq 0$ and $t\geq 0$.

(b) **(After the lecture on Friday)** Evaluate the flux integral $\iint_{\mathcal{S}} \vec{F}\cdot\vec{n}dS$ where $\vec{F}(x,y,z)=(z,x,y^2)$ and $\mathcal{S}$ is the part of the paraboloid $z=x^2+y^2$ lying inside the cylinder $x^2+y^2=4$ and whose boundary is oriented counterclockwise.
Hint: What is the vector function describing $\mathcal{S}$?

5. **(After the lecture on Friday)** We recall here the curl and divergence of a vecor field:

$$\boxed{\begin{aligned} \mathrm{curl}(\vec{F}) &= \nabla\times\vec{F} = \left(\frac{\partial R}{\partial y}-\frac{\partial Q}{\partial z},\frac{\partial P}{\partial z}-\frac{\partial R}{\partial x},\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right) \\ \mathrm{div}(\vec{F}) &= \nabla\cdot\vec{F} = \frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z} \end{aligned}}$$

If $\vec{F}(x,y,z):=(2x^2-yz,xyz^2,-x-2zy^2)$ then find $\mathrm{div}(\vec{F})$ and $\mathrm{curl}(\vec{F})$.

6. **(After the lecture on Friday)**We recall here the Stokes theorem:
Let $\mathcal{S}$ be a piecewise smooth, oriented surface, whose oriented boundary $\mathcal{C}$ is a piecewise smooth, simple closed curve. If $\vec{F}=(P,Q,R)$ is a smooth, three-dimensional vector field whose domain is a region in $\mathbb{R}^3$ that contains the surface $\mathcal{S}$, then

$$\boxed{\oint_{\mathcal{C}} \vec{F}(x,y,z)\cdot d\vec{r} = \iint_{\mathcal{S}} \mathrm{curl}(\vec{F})(x,y,z)\cdot\vec{n}(x,y,z)\,dS}\,.$$

(a) Verify the Stokes theorem for the vector field $\vec{F}(x,y,z)=(y^2,z^2,y^2)$ and $S$ being the part of the plane $x+2y+3z=6$ lying in the first octant.

(b) Verify the Stokes theorem for Question 3(b).